

Adaptive TX FFE with Sign-Sign LMS: a Symbol-Rate Simulation on a Backplane Channel

Goal

Reproduce the dual-loop TX FFE adaptation of Stojanović et al. (JSSC 2005) in simulation: a 4-tap transmit FFE adapted by sign-sign LMS from a single receiver-side error sampler, over a real backplane channel (ECEN720 T20 server thru, 10 Gb/s NRZ). The unequalized channel loses 27.6 dB at Nyquist and its worst-case (peak-distortion) eye is closed.

System

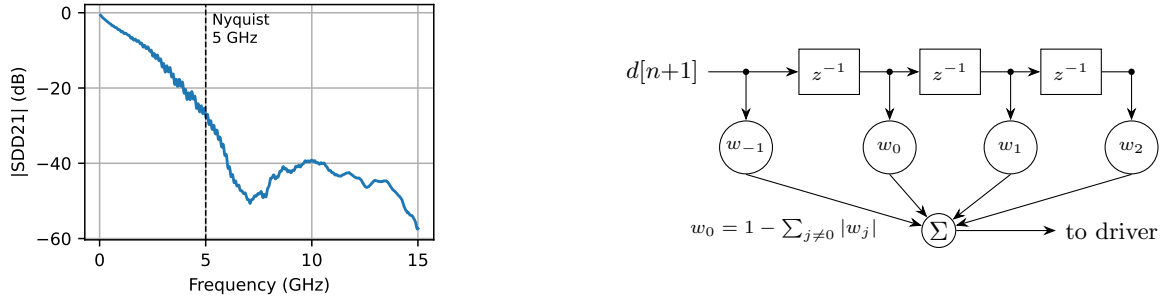


Figure 1: Left: channel loss. Right: symbol-spaced TX FFE (one pre-tap, two post-taps). The TX peak swing is fixed, so the main tap takes whatever the other taps leave: $\sum_j |w_j| = 1$.

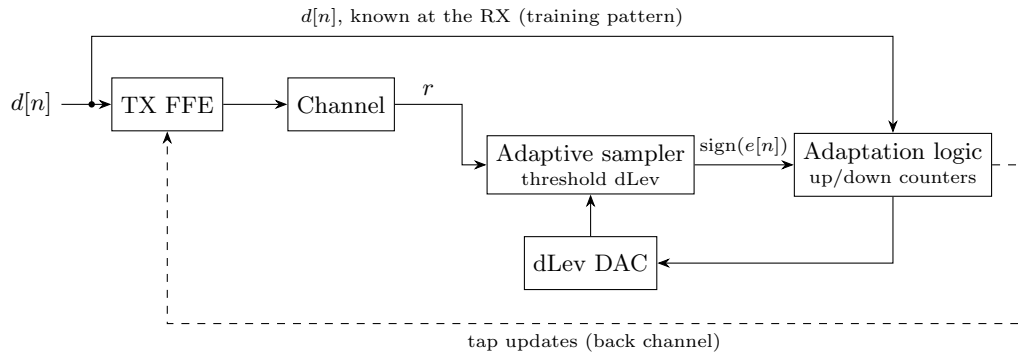


Figure 2: Adaptation loop as simulated. One extra comparator, whose threshold is the dLev DAC, gives $\text{sign}(e)$ directly; no ADC is needed. The data correlated against $\text{sign}(e)$ is the transmitted $d[n]$, i.e. a known training pattern (or error-free RX decisions).

Algorithm

With $e[n] = \text{dLev} - r[n]$, the two loops of the paper are

$$w_j \leftarrow w_j + \Delta_w \text{sign}\left(\sum_n \text{sign}(e[n]) d[n-j]\right), \quad j \neq 0, \quad \text{dLev} \leftarrow \text{dLev} - \Delta_{\text{dLev}} \text{sign}\left(\sum_n \text{sign}(e[n])\right),$$

summed over a block of symbols (one step per block, as an up/down counter would).

- **One sampler.** It sits at +dLev, so only symbols with $d[n] = +1$ contribute.
- **Main tap from the constraint.** Scaling all taps and dLev together leaves every $\text{sign}(e)$ unchanged, so the two loops alone cannot fix the overall scale (and can drift to all-zero). Setting $w_0 = 1 - \sum_{j \neq 0} |w_j|$ removes that freedom. dLev cannot be found first: it is the equalized main cursor, which depends on the final taps.
- **Averaged, it is zero-forcing.** $E[e[n]d[n-j]]$ is the residual ISI at cursor j , so each tap drives its own cursor to zero, even when the unequalized eye is closed.
- **Noise helps.** Without sampler noise the residual ISI takes few discrete values and $\text{sign}(e)$ has a dead zone around the solution (tap error $\sim 5\times$ larger).

Simulation

Symbol-rate model in Python: the channel is its pulse response sampled once per UI at the peak (3 pre-, 40 post-cursors); TX FFE and channel run as a continuous stream across blocks. 3000 blocks $\times$ 512 symbols, $\Delta_w = \Delta_{\text{dLev}} = 10^{-3}$, 5 mV rms sampler noise. The loop uses the transmitted symbols, aligned to each received sample, in place of RX decisions. Eyes are then checked with a waveform-level link (32 samples/UI, full S-parameter channel).

Results

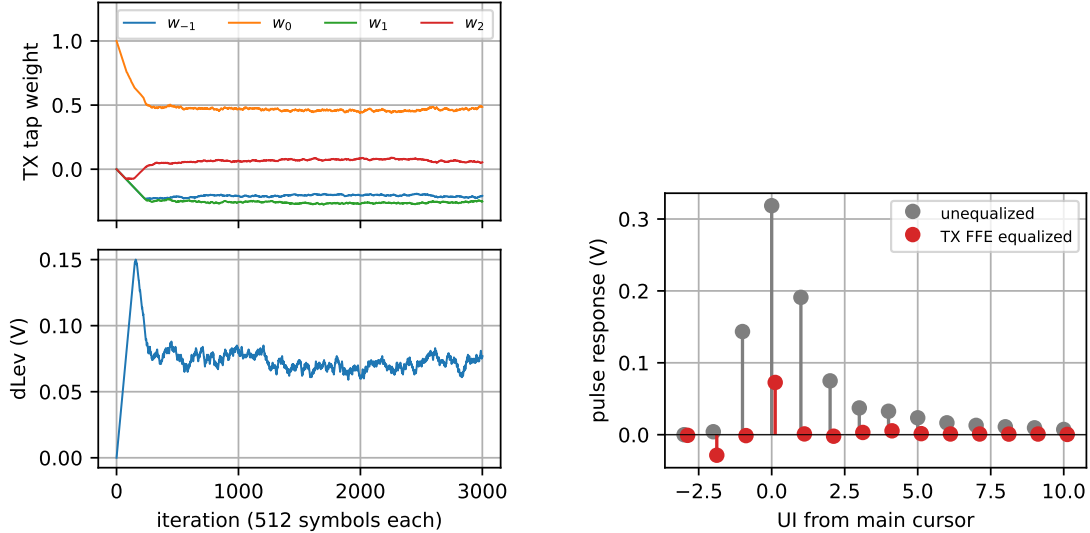


Figure 3: Left: taps and dLev converge together in ~ 300 blocks; dLev overshoots, then follows the main cursor down as pre-emphasis takes swing. Right: UI-spaced pulse response; the cursors within the FFE span (-1 , $+1$, $+2$) are driven to zero.

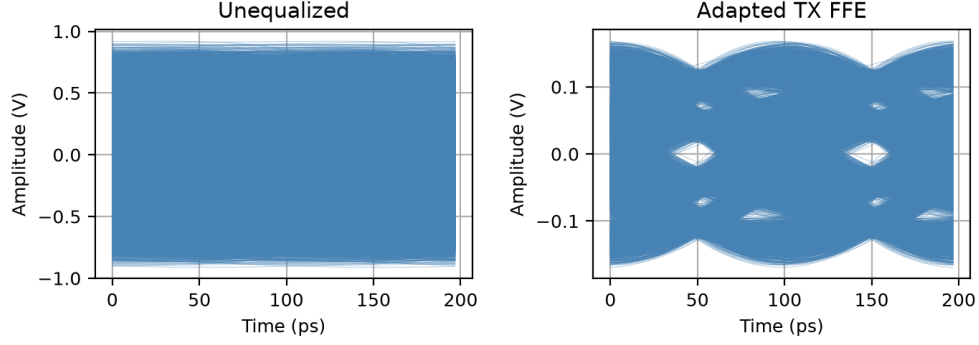


Figure 4: 10 000 random bits through the waveform-level link, before and after adaptation.

	main cursor	$\sum ISI $	PDA eye height
Unequalized	0.319 V	0.630 V	-0.622 V (closed)
Adapted TX FFE	0.073 V	0.061 V	+0.024 V (open)

Table 1: Adapted taps $[w_{-1}, w_0, w_1, w_2] = [-0.212, 0.470, -0.256, 0.061]$ (mean of the last 500 blocks); the exact zero-forcing solution is $[-0.205, 0.462, -0.261, 0.072]$.

Takeaways

- The loop finds the zero-forcing taps blind, from one comparator, and opens a closed eye.
- The cost is swing: the received main cursor drops $4.4\times$, so the eye barely opens. Residual ISI is now outside the FFE span (a new pre-2 cursor and the long post-cursor tail), which points to adding a DFE next (the paper’s loop-unrolled 1-tap DFE).
- At this low SNR, sign-sign keeps wandering slowly around the solution; results are averaged.

Adding a 1-tap DFE

The TX FFE pays for every cursor it cancels with swing: its main tap fell to 0.47, so the received main cursor dropped 4.4 $\times$. A receiver DFE subtracts the first post-cursor using past decisions instead, at no TX swing cost. Configuration as in the paper: the same 4-tap TX FIR plus a 1-tap DFE, with all loops adapted by sign-sign LMS. (The paper finds the DFE tap by locking dLev to two signal levels; here it is adapted directly.)

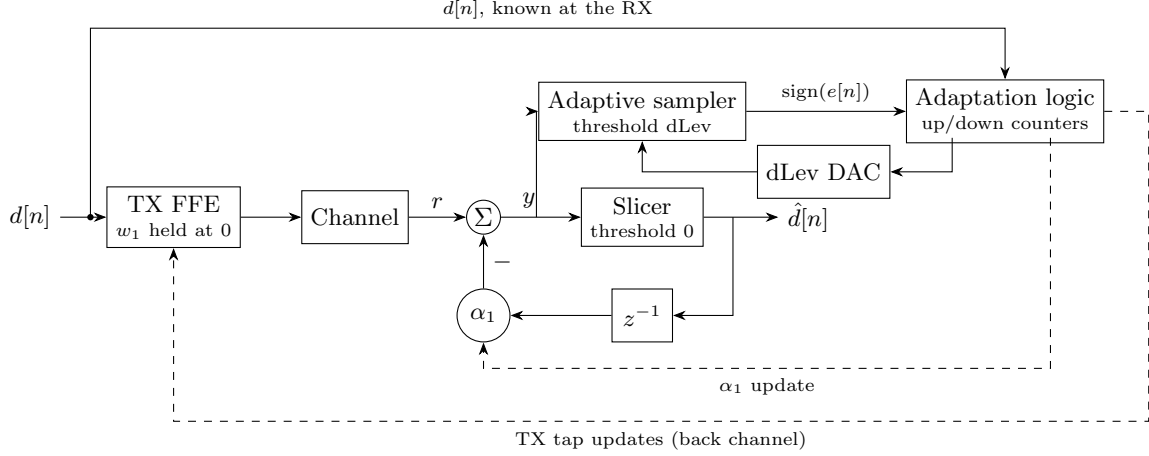


Figure 5: TX FFE + 1-tap DFE. The adaptive sampler sees the DFE summer output y ; the same $\text{sign}(e)$ updates the TX taps, α_1 and dLev (through the dLev DAC).

With $e[n] = \text{dLev} - y[n]$, the DFE tap adds one more counter:

$$\alpha_k \leftarrow \alpha_k - \Delta_\alpha \text{sign}\left(\sum_n \text{sign}(e[n]) d[n - k]\right) \quad (\text{minus: the DFE subtracts}).$$

- **One knob per cursor.** Averaged, w_j drives p_j to 0, α_k tracks p_k , and dLev tracks p_0 . A TX post-tap at a cursor the DFE covers is driven by the same $p_k - \alpha_k$ as α_k , so the split between them would be undetermined: TX post1 is held at 0.
- **All loops together.** The TX equations don't depend on α or dLev; those only track cursors the TX sets, so they follow it as it converges.

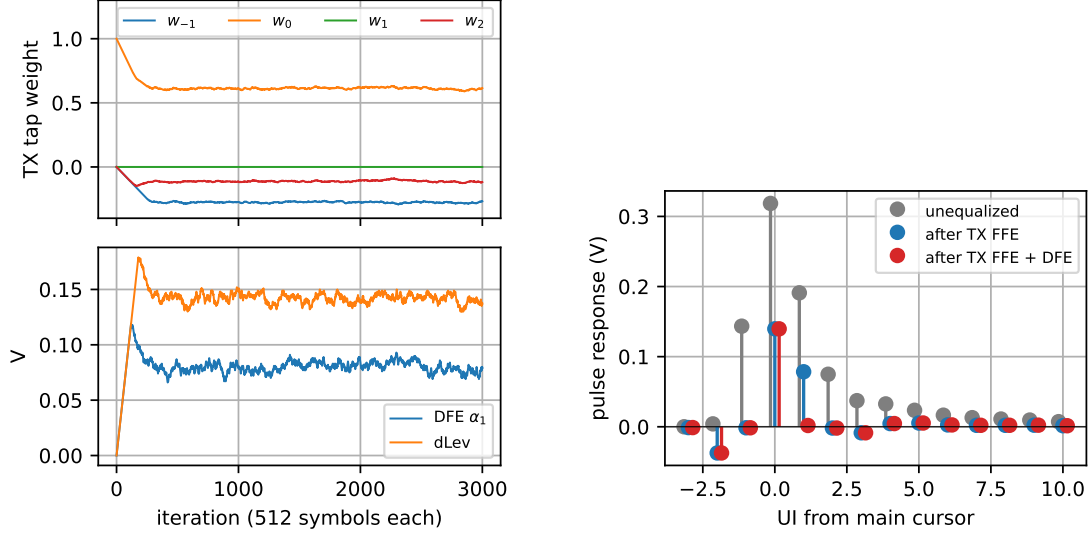


Figure 6: Left: TX post1 (w_1) stays at 0 and the main tap settles near 0.6; α_1 and dLev converge with the TX taps. Right: after the TX FFE the first post-cursor remains (blue); the DFE removes it (red), and the main cursor is about twice the FFE-only one.

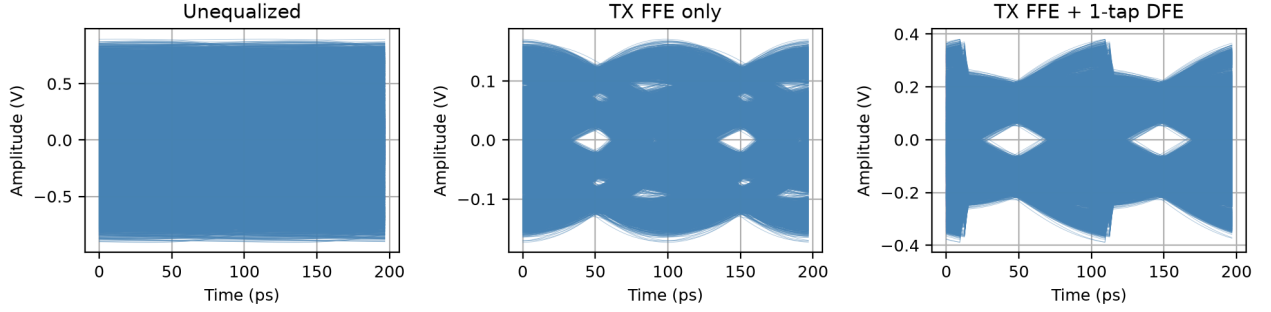


Figure 7: Waveform-level eyes, 10000 random bits. The DFE decides at the pulse-peak instant from its own past decisions; its feedback is held over the UI centered on that instant (hence the steps away from the eye center).

	TX taps $[w_{-1}, w_0, w_1, w_2]$	α_1	main cursor	PDA eye height
Unequalized			0.319 V	-0.622 V
TX FFE only	$[-0.212, 0.470, -0.256, 0.061]$		0.073 V	+0.024 V
TX FFE + 1-tap DFE	$[-0.278, 0.606, 0, -0.116]$	0.077 V	0.140 V	+0.091 V

Table 2: Adapted settings (mean of the last 500 blocks) and the resulting worst-case eye.

Takeaways. The first DFE tap gives most of the gain ($3.8\times$ the PDA eye): it takes the largest post-cursor without spending swing, so the main tap returns to 0.61. The TX still has to cancel the pre-cursor (its pre-tap grows to -0.28) and now also post2. In a sweep, more DFE taps saturate near $+0.12$ V.